

## Ideation Phase

### Define the Problem Statements

|                      |                                                |
|----------------------|------------------------------------------------|
| <b>Date</b>          | 2 August 2026                                  |
| <b>Team ID</b>       | Not specified                                  |
| <b>Project Name</b>  | Digitize Metro Ticket Booking Using ServiceNow |
| <b>Maximum Marks</b> | 2 Marks                                        |

#### Customer Problem Statement Template:

The Metro Ticket Generating System is designed to solve the challenges faced by metro passengers and metro authorities during the ticket booking process. In many metro stations, passengers experience long waiting times due to manual ticket booking, especially during peak hours. Paper-based tickets can easily be lost, damaged, or misplaced, creating inconvenience during travel. In addition, passengers often find it difficult to know the exact fare between stations before purchasing a ticket.

From the perspective of metro administrators, manual ticket management increases workload and makes it difficult to monitor ticket sales, generate reports, and analyze passenger travel patterns. A lack of automation also increases the possibility of human errors and delays.

The proposed Metro Ticket Generating System aims to provide a digital platform built on the ServiceNow platform that enables passengers to book tickets online, automatically calculates fares, generates QR-code-based tickets, and stores booking history securely. The system also provides administrators with automated reports and dashboards to improve operational efficiency.

#### Objectives

The objectives of the project are:

- Digitize metro ticket booking
- Reduce waiting time
- Eliminate paper tickets
- Generate secure QR-code tickets
- Automate fare calculation
- Improve passenger convenience
- Reduce manual work
- Improve reporting and analytics
- Enhance operational efficiency
- Increase customer satisfaction

## Problem Statements — Smart Metro Ticket Generation System

| Problem Statement (PS) | I am (Customer)                                                                                | I'm trying to                                                        | But                                                                                                   | Because                                                                                                       | Which makes me feel     |
|------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------|
| PS-1                   | a daily working commuter who travels by metro every weekday during peak hours                  | buy a ticket quickly and board the train without missing it          | the ticket counters have long queues and vending machines often run out of change or are out of order | there are a limited number of counters/machines and the process still relies heavily on cash and exact change | Frustrated and stressed |
| PS-2                   | an occasional traveler or tourist who is new to the city and unfamiliar with the metro network | find the correct route and fare to reach my destination without help | the ticket vending interface is confusing and doesn't clearly explain routes or fares in my language  | the system is not intuitive and offers little multilingual or first-time-user guidance                        | Confused and anxious    |
| PS-3                   | a senior citizen who is not comfortable using smartphones or digital payment apps              | purchase a ticket on my own without relying on someone else          | most modern ticketing options assume everyone has a smartphone and digital wallet                     | the system wasn't designed with non-digital-native or elderly users in mind                                   | Excluded and dependent  |

### Root Causes

The major reasons for the existing problems are:

- Manual ticket booking process
- Long queues during peak hours
- Lack of online ticket booking
- Absence of automatic fare calculation
- Dependence on paper tickets
- No QR-code-based ticket verification
- No centralized booking database
- Limited reporting and analytics
- Higher operational workload
- Human errors during ticket management

### Stakeholders

The following stakeholders are involved in the project:

- Metro Passengers
- Students
- Daily Office Employees
- Tourists
- Metro Ticket Operators
- Metro Administrators

- Metro Management Team
- ServiceNow Administrators

### **Existing System Limitations**

The existing metro ticketing process has several drawbacks:

- Time-consuming booking process
- Long waiting queues
- Manual fare calculation
- Paper ticket dependency
- No digital booking history
- Limited automation
- Difficult report generation
- Poor passenger convenience
- Increased operational costs

### **Conclusion**

The **Metro Ticket Generating System** provides a modern and efficient solution for metro ticket booking by leveraging the capabilities of the ServiceNow platform. It eliminates manual ticketing challenges through online booking, QR-code generation, automated fare calculation, and workflow automation. The project enhances both passenger convenience and administrative efficiency while supporting the digital transformation of public transportation. It offers a scalable, secure, and user-friendly solution that improves service quality and contributes to a smarter metro ecosystem.